



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Discontinuity Report No. 2025 – 0017



Agricultural Machinery – Internal Combustion Engine YANMAR TF70L

Agricultural Machinery	Internal Combustion Engine
Machine Brand and Model	YANMAR TF70L
Serial Number	320226E
Date issued	January 15, 2026
Test location	Roberto C. Bautista Hall, UPLB, College, Los Baños, Laguna
Items tested	Maximum Power Test
Test Applicant	ALL CERTIFIED EQUIPMENT TRADING CORP. 905 EDSA Brgy. Philam Homes, Quezon City

Summary of Incident

Date of Incident	04 August 2025
Time of Incident	01:00 p.m.
Type of Incident	The test applicant voluntarily terminated the maximum power test after evaluating the measured power output and fuel consumption, as the target performance values were not achieved; consequently, the succeeding tests were also discontinued.
Description of Incident	The YANMAR TF70L Internal Combustion Engine was tested based on PNS/BAFS 397:2024 Agricultural Machinery – Internal Combustion Engine – Methods of Test. The performance test started on the maximum power test on which the engine will be subjected to a load where maximum power occurs for two hours (If maximum power occurs at the given rated speed). During the test, maximum power was measured with a value of 4.72 kW occurred at 2401 rpm. At its 120th minutes of reading, the client decided to voluntarily terminate the test after observing its power output, and fuel consumption.

SUMMARY

Performance Criteria	Requirement as per PNS/BAFS 396:2024	AMTEC Test
Maximum Power as Percentage of Rated Maximum Power, %, minimum	80	NM ^a
Average specific fuel consumption during the continuous running test, g/kWh, maximum	350	NM ^b
Rated maximum power at rated speed, kW	5.22 kW at 2400 rpm ^c	4.72 kW at 2401 rpm ^d
Rated continuous power at rated speed, kW	4.50 kW at 2400 rpm ^c	NM ^a
No major breakdowns/ malfunctions during the test	None	NM ^e

^a Varying load was not conducted.

^b Continuous running test was not conducted.

^c Based on the nameplate provided by the test applicant.

^d Average of 7 readings during the maximum power test.

^e The test applicant voluntarily terminated the ongoing maximum power test after evaluating the measured power output and fuel consumption, as the target performance values were not achieved; consequently, the succeeding tests were also discontinued

Note: NM – Not Measured

RESULTS

Maximum Power Test

Table 1. Test conditions during the maximum power test of YANMAR TF70L Diesel Engine.

Date of test	August 04, 2025
Ambient temperature, °C	34.3
Relative humidity, %	65.2
Atmospheric pressure, mb	1019.9

Table 2. Results of maximum power test of the YANMAR TF70L Diesel Engine.

TIME (mins)	ENGINE SPEED (rpm)	SHAFT TORQUE (N-m)	OUTPUT POWER (kW)	FUEL CONSUMPTION		TEMPERATURE (°C)			
				HOURLY (L/h)	SPECIFIC (g/kWh)	FUEL	ENGINE OIL	COOLANT	EXHAUST
0	2399	19.45	4.89	1.94	333.51	34.3	91.6	100.9	302.1
20	2335	18.93	4.76	1.61	283.97	35.5	109.8	113.7	353.7
40	2401	18.53	4.66	1.63	293.79	36.2	115.8	118.1	389.3
60	2401	18.50	4.65	1.58	285.37	36.8	116.3	118.2	389.9
80	2400	18.77	4.72	1.64	292.08	36.9	116.0	118.2	421.2
100	2401	18.57	4.67	1.69	304.10	37.6	116.4	118.4	420.1
120	2401	18.60	4.68	1.68	301.76	37.7	116.2	118.1	431.1
AVE	2401	18.76	4.72	1.68	299.22	36.43	111.73	115.09	386.77
Standard deviation			0.087						
Percent Variation			1.78						

Summary of Incident (Cont'n)

Date of Incident	04 August 2025
Time of Incident	03:30 p.m.
Type of Incident	The test applicant voluntarily terminated the ongoing maximum power test after evaluating the measured power output and fuel consumption, as the target performance values were not achieved; consequently, the succeeding tests were also discontinued.
Description of Incident	The YANMAR TF70L Internal Combustion Engine was tested based on PNS/BAFS 397:2024 Agricultural Machinery – Internal Combustion Engine – Methods of Test. The performance test started on the maximum power test on which the engine will be subjected to a load where maximum power occurs for two hours (If maximum power occurs at the given rated speed). During the test, maximum power was measured with a value of 4.78 kW occurred at 2200 rpm. Since the maximum power did not occur at the given rated speed, supposedly, an additional one-hour was carried out. At its 80th minutes of reading, the client decided to voluntarily terminate the ongoing test after observing its power output, and fuel consumption.

SUMMARY (Cont'n)

Performance Criteria	Requirement as per PNS/BAFS 396:2024	AMTEC Test
Maximum Power as Percentage of Rated Maximum Power, %, minimum	80	NM ^a
Average specific fuel consumption during the continuous running test, g/kWh, maximum	350	NM ^b
Rated maximum power at rated speed, kW	5.22 kW at 2400 rpm ^c	4.31 kW at 2200 rpm ^d
Rated continuous power at rated speed, kW	4.50 kW at 2400 rpm ^c	NM ^a
No major breakdowns/ malfunctions during the test	None	NM ^e

^a Varying load was not conducted.

^b Continuous running test was not conducted.

^c Based on the nameplate provided by the test applicant.

^d Average of 5 readings during the maximum power test.

^e The test applicant voluntarily terminated the ongoing maximum power test after evaluating the measured power output and fuel consumption, as the target performance values were not achieved; consequently, the succeeding tests were also discontinued

Note: NM – Not Measured

RESULTS (Cont'n)

Maximum Power Test and Power at rated engine speed test

Table 3. Test conditions during the maximum power test of YANMAR TF70L Diesel Engine.

Date of test	August 4, 2025
Ambient temperature, °C	29.9
Relative humidity, %	85.8
Atmospheric pressure, mb	1011.9

Table 4. Results of maximum power test of the YANMAR TF70L Diesel Engine.

TIME (mins)	ENGINE SPEED (rpm)	SHAFT TORQUE (N-m)	OUTPUT POWER (kW)	FUEL CONSUMPTION		TEMPERATURE (°C)			
				HOURLY (L/h)	SPECIFIC (g/kWh)	FUEL	ENGINE OIL	COOLANT	EXHAUST
0	2200	18.63	4.29	1.89	369.83	30.7	80.9	106.1	495.8
20	2200	18.67	4.30	1.87	365.26	31.5	110.0	108.9	523.6
40	2200	18.83	4.34	1.90	367.89	31.6	124.6	109.0	560.9
60	2200	18.67	4.30	1.87	365.26	31.6	137.7	109.2	572.5
80 ^a	2201	18.63	4.29	1.86	363.85	32.1	140.3	109.2	602.7
AVE	2200	18.69	4.31	1.88	366.42	31.47	118.70	108.48	551.10
Standard deviation			0.019						
Percent Variation			0.44						

^a The maximum power did not occur at 2400 rpm rated engine speed. The test applicant voluntarily terminated the ongoing maximum power test after evaluating the measured power output and fuel consumption, as the target performance values were not achieved; consequently, the succeeding tests were also discontinued.

OBSERVATIONS

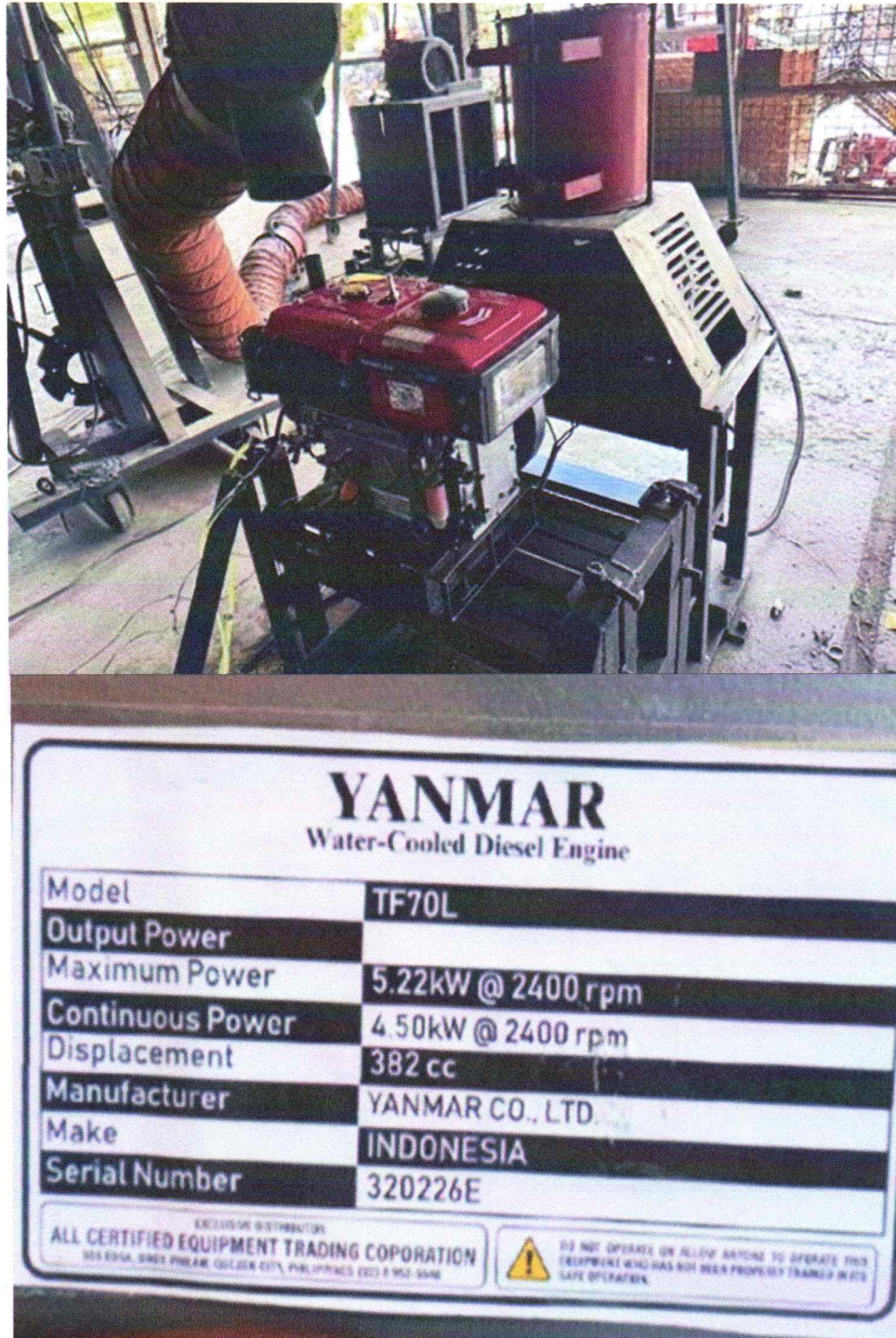


Figure 1. Engine test set up and nameplate of the YANMAR TF70L Diesel Engine.

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